

RIDDHI RAINA PRASAD

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SUMMARY

Software engineer with full-stack development experience and a strong analytical foundation, passionate about building reliable, product-facing systems. Experienced in shipping innovative technical solutions end to end.

PROFESSIONAL EXPERIENCE

Founder, Ally | NYC, NY

Apr '26 - Present

(AI Accessibility Tool | Base44 Inaugural Accelerator, 1 of 33 ventures selected)

- Developed and shipped a working AI-powered product in under **48** hours, winning **1st place** at Cross-Campus 44 Hackathon, by integrating five live external APIs into a single **LLM orchestration** layer with Twilio, Google Calendar, Gmail, and Maps.
- Improved daily accessibility outcomes for differently-abled students, **selected for Base44's inaugural accelerator cohort**, by building an end-to-end event-driven integration layer with proactive SMS alerts and accessible-route navigation.

Graduate Researcher, Data Systems & ML, New York University | NYC, NY

Apr '26 - Present

(National Grid x NYU: High-Resolution Energy Demand Dataset)

- Built a Python data pipeline integrating NYC building records, utility data, and energy simulation outputs, improving dataset accessibility for **5-borough** infrastructure planning. Structured and cleaned heterogeneous urban data sources for ML model training.
- Designed and evaluated modeling and simulation workflows using **XGBoost** to estimate daily building-level natural gas consumption, improving forecast accuracy for time-series demand patterns, by engineering features from building characteristics data.

Software Engineer II, UnitedHealth Group - Optum | Hyderabad, India

Mar'25 - Aug '25

- Built a GPT-4 agentic **AI code review tool** on Azure OpenAI that automated multi-step diff analysis and posted structured inline feedback to pull requests, reducing **CI/CD** overhead by **40%** and saving **12+** engineering hours per week.
- Collaborated cross-functionally with **8** global engineering teams to standardize release workflows and drive adoption, cutting review turnaround from days to minutes across distributed development environments.

Software Engineer, UnitedHealth Group - Optum | Hyderabad, India

Jun'22 - Mar'25

- Rebuilt a legacy web application into a **React TypeScript** single-page application, improving workflow usability for **5,000+ daily users** across **4 product lines**, by migrating from Bootstrap to a modern component-based architecture with a .NET Core backend.
- Migrated legacy auth to **Zero-Trust OAuth 2.0** and JWT across **12** clinical apps, securing **1M+** HIPAA-compliant patient records and eliminating **15+ SQL injection** vulnerabilities. Enforced token validation and role-based access controls.
- Automated **40+ regression test** workflows with Playwright and Selenium, reducing QA cycle time by **50%** and catching pre-production defects earlier, by integrating **E2E test suites** directly into **GitHub Actions** CI pipelines.
- Reduced **p95** API response time by **35%** across distributed .NET and React services, improving performance for **2,000+** internal users, by introducing **Redis** caching, **gRPC** inter-service communication, and async connection pooling.

PROJECTS

NYC Transit Hub | Python, Flask, React, SQLite, Docker

Apr '26

- Aggregated live **MTA** bus and subway data from **13 GTFS-RT** Protobuf feeds into a unified **REST API**, serving real-time arrivals and alerts for **1,488** stops across **13** endpoints.
- Built **React** frontend with **O(log n)** indexed lookups and Docker deployment. **Won Best Project at NYU.**

Backsie: Multi-Agent Community AI Platform | Next.js, FastAPI, PostgreSQL, Redis, Docker

Jan '26

- Built a community app that understands emotional depth, routing users through context-aware conversation flows that detect sentiment, label emotional states, and respond with adaptive support. Shipped to production on **iOS** with real-time Server-Sent Events (**SSE**) streaming, **Redis** sessions, and **AWS Cognito auth**, serving **21+ active users**.

DocuFlow: Serverless Document Auditing Engine | Python, FastAPI, AWS Lambda, pgvector, React

Nov '25

- Built a serverless document auditing engine that ingests PDFs, chunks and indexes content using **pgvector** Hierarchical Navigable Small World (**HNSW**), and retrieves grounded compliance answers via **Gemini 1.5 Flash** at sub-100ms latency. Reduced manual audit review time by **60%**. Placed **2nd of 500+ teams** at **AWS Global Financial AI Hackathon**.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, TypeScript, C#, SQL, HTML/CSS

Database: PostgreSQL, Redis, SQLite, MongoDB

Libraries/Framework/Tools: React, Next.js, FastAPI, Flask, Node.js, .NET Core, Django, LangChain, LangGraph, pgvector, XGBoost, Docker, Kubernetes, GitHub Actions, Prometheus, Grafana

Cloud Technologies: Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP)

Development Practices: Full-Stack Development, CI/CD, Agile, Scrum, TDD, REST, gRPC, Microservices, OAuth 2.0, JWT, SOLID, Code Reviews

EDUCATION

Masters in Computer Science - New York University | 3.83/4.0

Aug '25 - May '27

- Teaching Assistant, Lean Launch Pad (NYU Courant) | K-12 STEM Course Instructor, ieSoSC (Innovation, Entrepreneurship, and the Science of Smart Cities)

Bachelors in Computer Engineering - Vellore Institute of Technology | 8.93/10

Jun '18 - Aug '22

Relevant Coursework: Software Engineering, Machine Learning, Big Data, Design and Analysis of Algorithms, Distributed Systems, Networks